Appendix

On the replicability of LLM-based linguistic annotation: Insights from close and extended replications of Morin & Marttinen Larsson (2025) using open-weight LLMs

Full results

Distribution of classifications in the testing set provided by Morin & Marttinen Larsson (2025)

Table 1: Distribution of classifications in the testing set compared to the human gold standard

Classification	n	Percentage
evaluative	39	38%
non-evaluative	63	62%
Total	102	100%

Close replication

The following tables represent the results of the close replication.

Accuracy of all models compared to the human gold standard

Table 2: Classification accuracy and agreement with the human gold standard

Model	Accuracy	Alignments
Claude	93%	95/102
GPT	94%	96/102
Mistral	67%	68/102

Accuracy as 2x2 matrices

Table 3: GPT classifications compared to the human gold standard

Table 3: Prediction (GPT) vs Reference (Human)

	evaluative	non-evaluative
evaluative	37	4
non-evaluative	2	59

Table 4: Mistral classifications compared to the human gold standard

Table 4: Prediction (Mistral) vs Reference (Human)

	evaluative	non-evaluative
evaluative	7	2
non-evaluative	32	61

Table 5: Claude classifications compared with the human gold standard

Table 5: Prediction (Claude) vs Reference (Human)

	evaluative	non-evaluative
evaluative	37	5
non-evaluative	2	58

Accuracy metrics for all models in close replication

Table 6: Classification performance of the three models (Mistral, GPT and Claude) compared to the human gold standard

Metric	Mistral	GPT	Claude (original study)
Accuracy	67%	94%	93%
Precision	77.8%	90.2%	88.1%
Recall	17.9%	94.9%	94.9%
F1-score	29.2%	92.5%	91.4%
Matthew's correlation coefficient	0.25	0.88	0.86

Extended replication with GPT: reduced training process

Accuracy compared to the human gold standard

Table 7: GPT performance in extended replication (reduced training: only pretraining dataset and testing)

Accuracy	Alignments
94%	96/102

Accuracy as 2x2 matrices

Table 8: GPT classifications with few-shot training compared to the human gold standard

Table 8: Prediction (GPT reduced training) vs Reference (Human)

	evaluative	non-evaluative
evaluative	39	6
non-evaluative	0	57

Accuracy metrics of GPT with reduced training compared to Claude in original study

Table 9: Classification performance of GPT after reduced training and Claude compared to the human gold standard

Metric	GPT (reduced training)	Claude (original study)
Accuracy	94%	93%
Precision	86.7%	88.1%
Recall	100%	94.9%
F1-score	92.9%	91.4%
Matthew's correlation coefficient	0.89	0.86

Further technical details on the study

The open LLMs were accessed remotely via an API. The generation parameters in the API call were set to *temperature* = 0, $top_p = 1$, and a constant random seed was used. Setting *temperature* to 0, meaning the model always chooses the most probable next token, removes random sampling and therefore makes additional constraints such as top_k (sampling only from the k most probable tokens) unnecessary and potentially confounding. Even though these measures were taken to improve reproducibility, identical outputs across runs cannot be guaranteed.